MIL-S-19500/167B(NAVY)
19 February 1968
SUPERSEDING
MIL-S-19500/167A(NAVY)
11 May 1964
(See 6.2)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICES, DIODE, TYPE 1N560

AND 1N561

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for 0.250 amp silicon power rectifiers and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Maximum ratings. -

Type	V_R	I_o	I_o at TA=150°C	if (surge) 1/120 sec.
	Vdc	Adc	Adc	A
1N560	800	0.75	0.250	25
1N561	1000	.75	.250	25

Operating temperature: - 65°C to + 175°C
Storage temperature: - 65°C to + 175°C
Barometric pressure, reduced:

1N560 30 mm Hg
1N561 54 mm Hg

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. - Diodes shall be in accordance with MIL-S-19500 and as specified herein.

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3.2 Abbreviations and symbols. - The abbreviations and symbol used herein are defined in MIL-S-19500.

3.3 Design, construction and physical dimensions. - Semiconductor diodes shall be of the design, construction and physical dimensions of JEDEC outline DO-1.

3.4 Performance characteristics. - Performance characteristics shall be as specified in 4.4 and 4.5.

3.5 Marking. - Diodes shall be marked in accordance with MIL-S-19500 except that the Country of origin and Manufacturer's identification may be omitted from the device. The polarity shall be indicated by conventional diode symbol. No color coding will be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests. - Qualification tests shall be conducted at a laboratory satisfactory to the Naval Electronics Systems Command. Qualification tests shall consist of the tests specified in 4.4 and 4.5. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification, SD -6 (see 6.1).

4.2 Quality conformance inspection. - Quality conformance inspection shall consist of the examinations and tests specified in 4.4 and 4.5.

4.3 Quality conformance inspection information. - When specified in the contract or order, one copy of the quality conformance inspection data pertinent to the inspection lot shall accompany the shipments.

4.4 Group A inspection. - Group A inspection shall consist of the examinations and test shown in table I.

4.5 Group B inspection. - Group B inspection shall consist of the examinations and tests shown in table II.

Table I -Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min.	Max.	Units
Subgroup 1			5				
Visual and mechanical examination	2071						
Subgroup 2			5				
Forward voltage drop	4011	I_F = 500 mAdc		V_F	---	1.1	Vdc
Reverse current	4016						
1N560		V_R = 800 Vdc		I_R		5	μAdc
1N561		V_R = 1000 Vdc		I_R	---	5	μAdc
Breakdown voltage	4021						
1N560		V_R = 1000 Vdc		I_R	---	100	μAdc
1N561		V_R = 1250 Vdc		I_R	---	100	μAdc
Subgroup 3			5				
High temperature operation		T_A = 175°C					
Reverse current	4016						
1N560		V_R = 800 Vdc		I_R	---	500	μAdc
1N561		V_R = 1000 Vdc		I_R	---	500	μAdc
Forward voltage, full cycle average	4011	T_A = 150°C I_O = 250 mAdc					
1N560		V= 560 Vac		V_F	---	0.5	V
1N561		V= 707 Vac		V_F	---	0.5	V
Reverse current, full cycle average resistive load	4016	T_A = 150°C I_O = 250 mAdc					
1N560		V= 560 Vac		I_R	---	200	μA
1N561		V= 707 Vac		I_R	---	200	μA

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Table II – Group B inspection

Examination or Test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min.	Max.	Units
Subgroup 1			20				
Physical dimensions	2066						
Subgroup 2			20				
Solderability	2026						
Thermal shock (temperature cycling)	1051	Test cond C, except T (high) = 175°C					
Thermal shock (glass strain)	1056 Condition A						
Moisture resistance	1021						
Subgroup 3			20				
Shock	2016	(nonoperating) 500G, 5 shocks, 1 msec 2 major axis					
Vibration fatigue	2046	(nonoperating) 10G					
Vibration, variable frequency	2056	(nonoperating) 10G					
Constant acceleration	2006	10,000G (non-operating) Orientations X_1, Y_1, Z_1					
Subgroup 4			20				
Terminal strength (tension)	2036 Condition A	4 lbs, 15 sec					
Terminal strength (lead fatigue)	2036 Condition E	4 arcs each lead					
Subgroup 5			20				
Barometric pressure, reduced	1001						
1N560		V_R = 800 Vdc 30mm Hg					
1N561		V_R = 1,000 Vdc, 54mm Hg					
Salt atmosphere	1041						
Subgroup 6			10				
Surge current	4066	T_A = 150°C. i(surge) = 25A I_F = 250 mAdc					
Endpoints (Subgroups 2, 3, 5 and 6)							
Forward voltage drop	4011	I_F = 500 mA		V_F	---	1.1	Vdc

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Table II -Group B inspection-Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min.	Max.	Units
Reverse current	4016						
1N560		V_R= 800 Vdc		I_R	---	5	μAdc
1N561		V_R= 1,000 Vdc		I_R	---	5	μAdc
Subgroup 7			λ = 10				
Steady state operation life	1026	T_A= 150°C. I_O= 250 mAdc f= 60 Hz					
1N560		V= 560 Vac					
1N561		V= 707 Vac					
Subgroup 8			λ = 10				
High temperature life (nonoperating)	1031	T_A= 175°C.					
Endpoints (Subgroups 7 and 8)							
Forward voltage drop	4011	I_F= 500mA		V_F	---	1.2	Vdc
Reverse current	4016						
1N560		V_R= 800 Vdc		I_R	---	10	μAdc
1N561		V_R= 1,000 Vdc		I_R	---	10	μAdc

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Naval Electronics Systems Command, Department of the Navy, Washington, D. C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". (Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activities:
Navy - SH, EC
User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC
(Project 5961-N051 (NAVY)

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